

HealthConnect Clinic – Week 6
Part 2 – TRACK SPECIFIC ASSESSMENT

Task 1 - Select Most Important Week 5 Findings for Deeper Investigation

Purpose

The purpose of this task is to identify the most important findings from Week 5 that require deeper investigation in Week 6. The selected findings were prioritised based on the strength of the observed relationship with appointment no-show behaviour and their potential to support business decisions and Data Science modelling.

1. Booking Lead Time and No-Show Behaviour

Booking lead time was selected as the highest-priority finding because it showed the clearest and strongest pattern from the Week 5 analysis.

The no-show rate increased as the number of days between booking and the appointment increased.

Booking Lead Group	No-Show Rate
0-9 days	29.01%
10-19 days	37.11%
20-29 days	44.44%
30-39 days	50.53%
40-49 days	61.62%
50-59 days	67.92%
60+ days	68.09%

Why this requires deeper investigation

This pattern could help identify when patients become more likely to miss their appointments. Further analysis can determine whether the relationship remains consistent across different patient or appointment segments.

The finding may also support practical interventions such as targeted reminders or changes to appointment scheduling.

The 60+ days group should be interpreted cautiously because it contains relatively few records.

2. Previous No-Shows and Future No-Show Behaviour

Previous no-show history was selected because the Week 5 analysis showed a strong relationship between a patient's previous no-shows and their likelihood of missing another appointment.

Previous No-Shows	No-Show Rate
0	43.52%
1	53.49%
2	59.36%
3	67.95%
4	66.67%

Why this requires deeper investigation

The results suggest that patients with a history of missing appointments may be more likely to miss future appointments.

Further investigation can determine whether this pattern remains meaningful across different patient and appointment segments.

This finding could also help identify patients who may benefit from more targeted follow-up or additional support.

The higher previous-no-show groups are smaller, so their results should be validated carefully before making strong conclusions.

3. Reminder Status and No-Show Behaviour

Reminder status was selected because Week 5 showed a measurable difference in no-show rates between appointments where a reminder was sent and those where no reminder was sent.

Reminder Status	No-Show Rate
Reminder sent	47.36%
No reminder	51.39%

The difference between the two groups was 4.03 percentage points.

Why this requires deeper investigation

Appointments where a reminder was sent had a lower observed no-show rate than appointments without a reminder.

Further analysis is needed to understand whether this difference remains consistent across different appointment types, booking lead times, previous no-show histories and other relevant segments.

The Week 5 result shows an association, not proof that reminders directly caused the lower no-show rate.

Why These Three Findings Were Prioritised

These three findings were selected because:

They showed the clearest relationships with no-show behaviour in the Week 5 analysis.

They have the greatest potential to help explain higher-risk appointment behaviour.

They can support practical decisions around appointment scheduling, patient follow-up and reminders.

They are relevant to the Data Science track because they may help inform modelling decisions and feature consideration.

They provide clear areas for deeper validation rather than simply repeating the Week 5 exploratory analysis.

Findings Not Prioritised for Deeper Investigation at This Stage

Gender, appointment day, appointment time and waiting time were not prioritised for the first deeper investigation because they showed relatively limited variation in no-show rates during Week 5.

Distance to the clinic showed some increase in no-show rates at higher distances, but the highest-distance group was relatively small. Therefore, distance was treated as a secondary finding rather than one of the three main priorities.

Week 6 Focus From Task 1

The next stage of the analysis will investigate whether the three selected findings remain meaningful when examined in greater detail.

The deeper analysis will focus on:

Testing whether the identified relationships remain consistent.

Examining patterns across relevant patient and appointment segments.

Investigating relationships that may help explain no-show behaviour.

Identifying findings that can support specific business actions.

Providing relevant analytical evidence to support Data Science modelling decisions.

Task 2 - Additional Analysis: Booking Lead Time and No-Show Behaviour

The Week 5 finding on booking lead time was investigated further by reviewing the total number of appointments, number of no-shows and no-show rate for each booking lead group.

Booking Lead Group	Total Appointments	No-Shows	No-Show Rate
0-9 days	817	237	29.01%
10-19 days	865	321	37.11%
20-29 days	810	360	44.44%
30-39 days	843	426	50.53%
40-49 days	826	509	61.62%
50-59 days	745	506	67.92%

60+ days	94	64	68.09%
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Finding

The additional analysis confirmed a clear upward pattern in no-show rates as booking lead time increased.

The no-show rate increased from 29.01% for appointments booked 0-9 days in advance to 67.92% for appointments booked 50-59 days in advance. The 60+ day group recorded a 68.09% no-show rate.

Interpretation

The results support the Week 5 finding that longer booking lead times are associated with higher observed no-show rates.

The pattern remains clear across the larger booking lead groups, particularly from 20-29 days through 50-59 days.

The 60+ day group contains only 94 appointments, so its 68.09% rate should be interpreted with caution. However, the overall pattern is still strong across the larger groups.

Validation Outcome

The Week 5 finding remains meaningful after additional analysis and should be retained as an important factor for further investigation and decision support.

Previous No-Shows and Future No-Show Behaviour

The Week 5 finding on previous no-shows was investigated further by reviewing the total number of appointments, number of no-shows and no-show rate for each previous no-show group.

Previous No-Shows	Total Appointments	No-Shows	No-Show Rate
0	2,921	1,271	43.51%
1	1,548	828	53.49%
2	438	260	59.36%
3	78	53	67.95%
4	12	8	66.67%

5	3	3	100.00%
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Finding

The additional analysis confirms a clear relationship between previous no-show history and subsequent no-show behaviour.

The no-show rate increased from 43.51% among patients with no previous no-shows to 53.49% for patients with one previous no-show, 59.36% for two previous no-shows, and 67.95% for three previous no-shows.

The groups with four and five previous no-shows were very small, with only 12 and 3 appointments respectively, so their rates should be interpreted with caution.

Interpretation

The results support the Week 5 finding that patients with a history of previous no-shows tend to have higher observed no-show rates for subsequent appointments.

This pattern could be useful for identifying patients who may require additional follow-up or targeted support.

However, the analysis shows an association rather than proof that previous no-shows directly cause future no-shows.

Validation Outcome

The Week 5 finding remains meaningful after additional analysis and should be retained as an important factor for further investigation and decision support.

The very small groups with four and five previous no-shows should not be used as the main basis for decisions because their percentages are based on very few appointments.

Reminder Status and No-Show Behaviour

The Week 5 finding on reminder status was investigated further by reviewing the total number of appointments, number of no-shows and no-show rate for appointments with and without reminders.

Reminder Sent	Total Appointments	No-Shows	No-Show Rate
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No	1,366	702	51.39%
Yes	3,634	1,721	47.36%

Finding

The additional analysis confirms that appointments where a reminder was sent had a lower observed no-show rate than appointments where no reminder was sent.

The no-show rate was 47.36% for appointments with a reminder, compared with 51.39% for appointments without a reminder. This represents a difference of 4.03 percentage points.

Interpretation

The result supports the Week 5 finding that reminder status is associated with appointment no-show behaviour.

The difference suggests that reminders may be a useful area for further investigation and could support decisions around patient communication and appointment follow-up.

However, this analysis shows an association and does not prove that reminders directly caused the lower no-show rate. Other factors may also influence whether a patient attends an appointment.

Validation Outcome

The Week 5 finding remains meaningful after additional analysis and should be retained as an important factor for further investigation and decision support.

Task 3: Segment Analysis

Task objective: Investigate important patterns across relevant patient, appointment or operational segments.

Segment Analysis 1 – Booking Lead Group × Appointment Type

No-show rates were compared across booking lead groups and appointment types to determine whether the booking lead-time pattern observed in Week 5 remained consistent across appointment segments.

Booking Lead Group	Diagnostic	Follow-up	General Consultation	Specialist
0–9 days	32.47%	30.53%	26.85%	29.22%
10–19 days	27.18%	41.20%	38.36%	35.12%
20–29 days	52.75%	43.69%	42.70%	44.68%
30–39 days	52.53%	52.84%	46.70%	54.22%
40–49 days	53.33%	66.40%	62.69%	57.58%
50–59 days	77.27%	72.86%	61.90%	68.18%
60+ days	66.67%	68.57%	70.27%	57.14%
Total	49.75%	51.23%	46.64%	47.44%

Interpretation: Longer booking lead times are associated with higher no-show rates across all four appointment types. The highest values occur mainly in the longer lead-time groups. The 60+ day group should be interpreted cautiously because it contains relatively few appointments.

Conclusion: The booking lead-time pattern remains meaningful across appointment types, indicating that the relationship is not limited to one particular appointment category.

Segment Analysis 2 – Previous No-Shows × Appointment Type

No-show rates were compared across previous no-show history and appointment type to determine whether the previous no-show pattern persisted across different appointment categories.

Previous No-Shows	Diagnostic	Follow-up	General	Specialist
0	44.67%	47.29%	40.43%	43.89%
1	54.64%	53.56%	54.60%	49.81%
2	57.38%	65.19%	55.23%	60.00%
3	90.00%	64.29%	62.86%	68.42%
4	—	83.33%	100.00%	33.33%
5	—	—	100.00%	100.00%

Interpretation: No-show rates generally increase as previous no-show history increases across the appointment types. Patients with two or more previous no-shows generally show higher no-show rates than patients with no previous no-shows. Some values for four and five previous no-shows are based on very small groups and should therefore be treated cautiously.

Conclusion: The pattern of previous no-shows being associated with future no-shows persists across appointment types, strengthening its relevance as an analytical factor.

Segment Analysis 3 – Reminder Status × Appointment Type

No-show rates were compared by reminder status and appointment type to determine whether the reminder pattern identified in Week 5 was consistent across appointment categories.

Reminder Status	Diagnostic	Follow-up	General	Specialist
No	55.56%	53.13%	49.16%	51.63%
Yes	47.88%	50.53%	45.65%	45.87%

Interpretation: The reminder group recorded a lower no-show rate across all four appointment types. The differences were 7.68 percentage points for Diagnostic Tests, 2.60 percentage points for Follow-up, 3.51 percentage points for General Consultation, and 5.76 percentage points for Specialist appointments.

Conclusion: The lower no-show rate among patients receiving reminders is consistent across appointment types. However, this is an association and does not establish that reminders directly caused patients to attend.

Overall Task 3 Finding

The segment analyses show that the main Week 5 patterns remain visible across relevant appointment segments. Longer booking lead times and previous no-show history are associated with higher no-show rates across appointment types, while reminder status is associated with lower no-show rates across the appointment categories examined. These findings provide stronger evidence for targeted patient-support decisions and future modelling.

Analytical caution: Small segments, particularly those involving four or five previous no-shows and the 60+ day booking lead group, can produce unstable rates. The results should therefore be used as directional evidence rather than definitive probabilities.

Task 4: Relationship Analysis

Task objective: Analyse relationships that could help explain appointment attendance or no-show behaviour.

Relationship 1 – Booking Lead Group × Previous No-Shows

This relationship was analysed to determine whether patients with a history of missed appointments become even more likely to miss future appointments when their appointments are booked further in advance.

Booking Lead Group	0 Previous	1 Previous	2 Previous	3 Previous	4 Previous	5 Previous
0–9 days	23.81%	33.05%	50.60%	17.65%	—	—
10–19 days	34.71%	36.91%	43.37%	83.33%	100.00%	100.00%
20–29 days	38.32%	49.64%	54.05%	84.62%	50.00%	—
30–39 days	46.98%	56.08%	56.92%	62.50%	—	—
40–49 days	55.77%	66.42%	77.27%	85.71%	100.00%	100.00%
50–59 days	61.94%	73.58%	82.26%	83.33%	100.00%	—
60+ days	64.71%	72.22%	60.00%	100.00%	—	—

Interpretation: The results show a generally increasing no-show pattern as booking lead time increases, particularly among patients with previous no-show history. For patients with 0 previous no-shows, the rate rises from 23.81% in the 0–9 day group to 64.71% in the 60+ day group. Patients with 1 previous no-show also show a generally increasing pattern, reaching 73.58% in the 50–59 day group and 72.22% in the 60+ day group.

Conclusion: Longer booking lead times and previous no-show history appear to combine into a higher-risk pattern for appointment non-attendance. Groups with three or more previous no-shows contain small numbers in some cells, so the 100% values should be interpreted cautiously.

Relationship 2 – Reminder Sent × Previous No-Shows

This relationship was analysed to determine whether no-show behaviour differs between patients who received reminders and those who did not, across different levels of previous no-show history.

Previous No-Shows	No Reminder	Reminder Sent
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0	45.44%	42.08%
1	56.54%	52.32%
2	66.67%	56.51%
3	70.00%	67.24%
4	100.00%	42.86%
5	—	100.00%

Interpretation: Patients who received reminders generally recorded lower no-show rates than patients who did not receive reminders across the groups with 0–3 previous no-shows. The largest difference among these main groups occurs for patients with two previous no-shows: 66.67% without a reminder compared with 56.51% with a reminder, a difference of 10.16 percentage points.

Conclusion: Reminder status and previous no-show history provide a useful relationship for understanding no-show behaviour. The results suggest that reminders may be useful when supporting patients with a history of missed appointments. The four- and five-previous-no-show groups are too small for strong conclusions, and the analysis shows association rather than causation.

Relationship 3 – Booking Lead Group × Reminder Sent

This relationship was analysed to determine whether the association between longer booking lead times and no-show behaviour differs depending on whether a reminder was sent.

Booking Lead Group	No Reminder	Reminder Sent
0–9 days	32.23%	27.89%
10–19 days	38.93%	36.39%
20–29 days	47.35%	43.32%
30–39 days	50.83%	50.42%
40–49 days	65.81%	59.97%
50–59 days	72.43%	66.43%
60+ days	87.50%	61.43%

Interpretation: No-show rates generally increase as booking lead time increases for both patients who received reminders and those who did not. The reminder group recorded a lower no-show rate in every booking lead category. The difference is particularly noticeable in the 40–49 day and 50–59 day groups, while the 60+ day group shows the largest difference: 87.50% without a reminder compared with 61.43% with a reminder.

Conclusion: The analysis suggests that longer booking lead times are associated with increased no-show behaviour, while reminders are associated with lower no-show rates within each booking lead group. This provides useful evidence for considering reminder strategies for appointments scheduled well in advance. The 60+ day result should be interpreted cautiously because it is a relatively small group, and the findings show association rather than causation.

Overall Task 4 Finding

Across the three relationship analyses, booking lead time and previous no-show history consistently emerge as important factors associated with future no-show behaviour. Reminder status also shows a generally lower no-show rate across relevant segments. Together, these relationships provide stronger decision-support evidence than examining the variables independently and can help inform targeted patient-support strategies and future predictive modelling.

Analytical caution: Some combinations, especially those involving four or five previous no-shows and the 60+ day booking lead group, contain relatively few appointments. These results should therefore be treated as directional evidence rather than definitive rates.

Task 5: KPI Validation

Objective

The purpose of this task was to validate the most important KPIs developed during Week 5 and confirm that the calculations remain accurate after the additional Week 6 analysis.

KPI Validation Results

KPI	Validated Result	Status
Total Appointments	5,000	Validated
No-Show Rate	48.46%	Validated
Attendance Rate	46.28%	Validated
Cancellation Rate	5.26%	Validated
Reminder No-Show Gap	4.03 percentage points	Validated

1. Total Appointments

The Total Appointments KPI was checked against the full HealthConnect appointment dataset.

The KPI returned 5,000 appointments, confirming that the total appointment count remains consistent with the Week 5 analysis.

Conclusion: The Total Appointments KPI remains accurate and valid.

2. No-Show Rate

The No-Show Rate KPI was validated using the total number of appointments and no-show appointments.

Total Appointments: 5,000

No-Shows: 2,423

No-Show Rate: 48.46%

Conclusion: The No-Show Rate remains accurate at 48.46% and is consistent with the Week 5 result.

3. Attendance Rate

The Attendance Rate KPI was checked against the total number of appointments and attended appointments.

Total Appointments: 5,000

Attended Appointments: 2,314

Attendance Rate: 46.28%

Conclusion: The Attendance Rate remains accurate at 46.28%.

4. Cancellation Rate

The Cancellation Rate KPI was validated using the total number of appointments and cancelled appointments.

Total Appointments: 5,000

Cancelled Appointments: 263

Cancellation Rate: 5.26%

Conclusion: The Cancellation Rate remains accurate at 5.26%.

5. Reminder No-Show Gap

The Reminder No-Show Gap was validated by comparing the no-show rates for appointments with and without reminders.

No-Show Rate With Reminder: 47.36%

No-Show Rate Without Reminder: 51.39%

Reminder No-Show Gap: 4.03 percentage points

This indicates that appointments with reminders had a lower observed no-show rate than appointments without reminders.

Conclusion: The Reminder No-Show Gap remains accurate at 4.03 percentage points. However, this represents an association and should not be interpreted as proof that reminders directly caused patients to attend.

Overall Task 5 Finding

The validation confirmed that all five selected Week 5 KPIs remain accurate and consistent during the Week 6 analysis. No changes to the KPI calculations were required.

The validated KPIs provide a reliable foundation for the deeper analysis and decision-support activities completed in Week 6.

Task 6: Business Impact of Key Findings

Objective

The purpose of this task was to identify which of the validated Week 6 findings have the greatest potential business impact for HealthConnect Clinic.

1. Booking Lead Time and No-Show Behaviour

Evidence:

The no-show rate increased from 29.01% for appointments booked 0-9 days in advance to 67.92% for appointments booked 50-59 days in advance. The relationship also remained visible across different appointment types and when analysed alongside reminder status.

Potential Business Impact: High

Longer booking lead times may help HealthConnect identify appointments that require additional attention before the appointment date. This could support targeted reminder and patient-support strategies for appointments scheduled further in advance.

2. Previous No-Shows and Future No-Show Behaviour

Evidence:

The no-show rate increased from 43.51% among patients with no previous no-shows to 67.95% among patients with three previous no-shows. The pattern also remained visible across different appointment types and when combined with booking lead time.

Potential Business Impact: High

Previous no-show history could help HealthConnect identify patients who may require additional support to improve appointment attendance. This finding may also be useful for future predictive modelling and risk identification.

3. Reminder Status and No-Show Behaviour

Evidence:

The no-show rate was 51.39% for appointments without a reminder compared with 47.36% for appointments with a reminder, giving a 4.03 percentage-point gap. The lower no-show rate associated with reminders was also observed across appointment types and booking lead groups.

Potential Business Impact: Medium-High

This finding provides a practical area for HealthConnect to consider when developing patient-support strategies. Strengthening reminder processes could potentially help reduce missed appointments, although the analysis shows an association and does not prove that reminders directly cause attendance.

Business Impact Ranking

Rank	Finding	Potential Business Impact
1	Booking lead time and no-show behavior	High
2	Previous no-shows and future no-show behaviour	High
3	Reminder status and no-show behaviour	Medium-High

Overall Task 6 Finding

The findings with the greatest potential business impact are booking lead time and previous no-show history, as both show strong and repeated associations with future no-show behaviour. Reminder status also has meaningful business potential because it represents an existing patient-support activity that HealthConnect can consider strengthening.

Together, these findings can support more targeted approaches to reducing missed appointments. However, the results represent associations rather than proof of causation, and small patient groups should be interpreted cautiously.

Task 7: Translating Findings into Specific HealthConnect Actions

Objective

The purpose of this task was to translate the significant analytical findings from Week 6 into specific actions that HealthConnect Clinic could consider to reduce missed appointments and improve patient support.

1. Booking Lead Time and No-Show Behaviour

Finding:

No-show rates increased as booking lead time increased. The no-show rate was 29.01% for appointments booked 0-9 days in advance and increased to 67.92% for appointments booked 50-59 days in advance.

Recommended Action:

HealthConnect should consider introducing stronger reminder and confirmation processes for appointments booked further in advance. Patients with longer booking lead times could receive an earlier reminder followed by a confirmation closer to the appointment date.

Expected Benefit:

This could help keep appointments visible to patients and provide an opportunity to identify patients who may need support before the appointment date.

Analytical Caution:

The analysis shows an association between longer booking lead time and higher no-show rates. It does not prove that longer lead time causes patients to miss appointments.

2. Previous No-Shows and Future No-Show Behaviour

Finding:

Patients with previous no-shows generally had higher future no-show rates. The rate increased from 43.51% among patients with no previous no-shows to 67.95% among those with three previous no-shows.

Recommended Action:

HealthConnect should consider using previous no-show history to identify patients who may benefit from additional appointment support. This could include targeted confirmation, additional reminders, or checking whether the patient needs assistance before the appointment.

Expected Benefit:

A more targeted support approach could allow the clinic to focus additional attention on patients who show a history of missed appointments rather than applying the same level of intervention to everyone.

Analytical Caution:

The relationship is based on historical appointment behaviour and should be used as a support signal rather than as a reason to make assumptions about individual patients.

3. Reminder Status and No-Show Behaviour

Finding:

The no-show rate was 47.36% for appointments where a reminder was sent, compared with 51.39% where no reminder was sent. This represents a difference of 4.03 percentage points.

Recommended Action:

HealthConnect should review and strengthen its reminder process to ensure that eligible patients receive timely reminders. The clinic could also monitor reminder coverage and identify situations where reminders are not being sent.

Expected Benefit:

Improving reminder coverage may support better patient awareness of upcoming appointments and provide patients with an opportunity to prepare or seek assistance before the appointment.

Analytical Caution:

The analysis shows an association between reminder status and no-show behaviour. It does not establish that reminders alone caused the lower no-show rate.

Overall Task 7 Recommendation

The analysis suggests that HealthConnect could use a targeted patient-support approach rather than relying on one general intervention for all appointments. Longer booking lead times and previous no-show history can be used as signals for additional attention, while the existing reminder process can be strengthened to support appointment attendance.

These actions should be monitored using the validated Week 6 KPIs to determine whether future changes are associated with improvements in attendance and reductions in no-shows.

Task 8: Cross-Track Analytical Contribution to Data Science

Objective

The purpose of this task was to provide relevant Data Analytics findings to the Data Science track and use the cross-track collaboration to support modelling decisions, validate analytical patterns and identify areas requiring further investigation.

Track Collaborated With

Data Science - Shylet

The Data Analytics findings were shared with the Data Science track following the Week 6 cross-track integration request.

1. Analytical Findings Provided to Data Science

The following validated Week 6 findings were shared:

Booking Lead Time

No-show rates increased substantially as booking lead time increased:

0-9 days: 29.01%

10-19 days: 37.11%

20-29 days: 44.44%

30-39 days: 50.53%

40-49 days: 61.62%

50-59 days: 67.92%

60+ days: 68.09%

This showed a strong relationship between longer booking lead times and higher no-show rates.

Previous No-Shows

No-show rates also increased with previous no-show history:

0 previous no-shows: 43.51%

1 previous no-show: 53.49%

2 previous no-shows: 59.36%

3 previous no-shows: 67.95%

The groups with 4 and 5 previous no-shows were very small and were therefore treated cautiously.

Appointment Type

Overall no-show rates by appointment type were:

Appointment Type	No-Show Rate
Diagnostic Test	49.75%
Follow-up	51.23%
General Consultation	46.64%
Specialist Consultation	47.44%

These results were shared to help Data Science compare the model's segment-level error patterns with the observed appointment behaviour.

Reminder Status

The analysis showed:

Without reminder: 51.39% no-show rate

With reminder: 47.36% no-show rate

Difference: 4.03 percentage points

This finding was also shared as a relevant patient-support signal, while maintaining the distinction between association and causation.

2. Data Science Feedback

The Data Science track reviewed the analytical findings against its model and error analysis.

The feedback confirmed that booking lead time was the strongest model feature, accounting for approximately 57% of the model's predictive weight. This aligned closely with the Data Analytics finding that no-show rates increased from 29.01% at 0-9 days to 67.92% at 50-59 days.

Data Science also reported that historical no-show rate was the second-strongest model feature at approximately 12%, which was consistent with the Data Analytics finding that no-show rates increased from 43.51% for patients with no previous no-shows to 67.95% for patients with three previous no-shows.

The reminder finding was also consistent with the Data Science review, with the approximately 4 percentage-point difference matching the pattern observed during the earlier analysis.

3. Clarification of Specialist Consultation Finding

The cross-track review identified an important distinction between model error rate and no-show rate.

Data Science initially identified Specialist Consultation as having the highest model error rate, at 42.2%. However, the Data Analytics findings showed that Specialist Consultation did not have the highest overall no-show rate. Follow-up had the highest overall no-show rate at 51.23%, followed by Diagnostic Test at 49.75%.

This clarification helped prevent the model's higher Specialist Consultation error rate from being interpreted as evidence that Specialist Consultation automatically represents the highest no-show-risk appointment type.

Data Science agreed to reframe this finding in its modelling write-up and recognised that the error pattern may be related to interactions between appointment type, booking lead time and previous no-show history.

4. Appointment Type and Booking Lead Time Interaction

The Data Analytics segment analysis showed that the relationship between booking lead time and no-show behaviour generally persisted across appointment types.

For example, at 0-9 days, the no-show rates were:

Diagnostic Test: 32.47%

Follow-up: 30.53%

General Consultation: 26.85%

Specialist Consultation: 29.22%

At 50-59 days, the rates increased to:

Diagnostic Test: 77.27%

Follow-up: 72.86%

General Consultation: 61.90%

Specialist Consultation: 68.18%

This supported further investigation of the interaction between appointment type and booking lead time.

The Data Science track indicated that this interaction could be considered when determining whether an explicit interaction feature should be tested during Week 7 modelling.

5. Information Received From Data Science

The Data Science track provided:

Model feature importance information.

Model error-analysis findings.

Confirmation that booking lead time was the strongest predictive feature.

Confirmation that historical no-show rate was another important predictive feature.

Clarification of the Specialist Consultation model-error finding.

A potential Week 7 modelling question involving appointment type and booking lead time.

6. How the Cross-Track Integration Supported the Project

The collaboration strengthened the HealthConnect analysis by providing an independent comparison between the Data Analytics findings and the Data Science model.

The alignment between the analytical findings and model feature importance increased confidence that booking lead time and previous no-show behaviour represent meaningful signals for understanding appointment no-shows.

The collaboration also identified an important modelling consideration: appointment type should not be interpreted in isolation, as its relationship with no-show behaviour may depend on other factors such as booking lead time and previous no-show history.

7. Integration Outcome

The cross-track collaboration resulted in the following outcomes:

1. Data Analytics findings were shared with Data Science.
2. Data Science reviewed the findings against model feature importance and error analysis.
3. Booking lead time was independently confirmed as the strongest model feature and aligned with the strongest Data Analytics finding.
4. Historical no-show behaviour was independently confirmed as an important model feature and aligned with the Data Analytics findings.
5. The Specialist Consultation model-error finding was clarified and reinterpreted correctly.
6. The interaction between appointment type and booking lead time was identified as an area for further Week 7 investigation.
7. The collaboration provided evidence that the Data Analytics work is contributing directly to modelling decisions and future testing.
8. Evidence of Integration

The following evidence supports the cross-track contribution:

Week 6 Data Analytics segment and relationship analysis.

Week 6 KPI validation.

Validated booking lead time, previous no-show and reminder findings.

Written Data Science feedback from Shylet.

Follow-up Data Analytics response addressing the Specialist Consultation finding.

Discussion of appointment type × booking lead time interaction for Week 7 modelling.

Cross-track coordination record documenting the Data Analytics → Data Science integration.

Conclusion

The Week 6 Data Analytics contribution was successfully connected to the Data Science track. The validated analytical findings were reviewed against the Data Science model and error analysis, with strong alignment observed for booking lead time and previous no-show behaviour.

The collaboration also improved the interpretation of the Specialist Consultation model error and identified appointment type × booking lead time as a potential interaction requiring further investigation during Week 7.

This integration demonstrates that the Data Analytics work is not operating independently but is directly supporting modelling interpretation, feature consideration and the next stage of HealthConnect development.

Task 9: Review and Revision of Week 5 Conclusions

Objective

The purpose of this task was to review the conclusions reached during Week 5 and determine whether the additional Week 6 analysis supports, refines, or changes those conclusions.

1. Booking Lead Time and No-Show Behaviour

Week 5 Conclusion:

Longer booking lead times were associated with higher no-show rates.

Week 6 Review:

The Week 6 analysis confirmed this finding. No-show rates increased from 29.01% for appointments booked 0-9 days in advance to 67.92% for appointments booked 50-59 days in advance. The relationship also remained visible across different appointment types and reminder statuses.

Revised Conclusion:

The Week 5 conclusion remains valid and is strengthened by the additional Week 6 analysis. Booking lead time remains one of the most important factors associated with no-show behaviour.

2. Previous No-Shows and Future No-Show Behaviour

Week 5 Conclusion:

Patients with a history of previous no-shows were more likely to have future no-shows.

Week 6 Review:

The Week 6 analysis confirmed that no-show rates generally increased as previous no-show history increased. The rate rose from 43.51% for patients with no previous no-shows to 67.95% for patients with three previous no-shows. The pattern was also observed across appointment types and when combined with booking lead time.

Revised Conclusion:

The Week 5 conclusion remains valid. Previous no-show history is an important behavioural indicator associated with future missed appointments.

3. Reminder Status and No-Show Behaviour

Week 5 Conclusion:

Appointments with reminders had a lower observed no-show rate than appointments without reminders.

Week 6 Review:

The Week 6 analysis confirmed this finding. The no-show rate was 47.36% for appointments with a reminder compared with 51.39% without a reminder, giving a difference of 4.03 percentage points. The lower rate with reminders was also observed across appointment types and booking lead-time groups.

Revised Conclusion:

The Week 5 conclusion remains valid. Reminder status continues to show an association with lower no-show rates, although the analysis does not establish that reminders directly cause the reduction.

Overall Revised Conclusion

The Week 6 analysis supports and strengthens the main conclusions from Week 5 rather than changing them.

The most important conclusions remain:

1. Longer booking lead times are associated with higher no-show rates.
2. Previous no-show history is associated with higher future no-show rates

3. Appointments with reminders have a lower observed no-show rate than appointments without reminders.

The Week 6 analysis provided additional segment and relationship analysis that increased confidence in these findings. However, associations should not be interpreted as proof of causation, and findings from very small groups should be treated cautiously.

Task 11: Evidence for Recommendations

Objective

The purpose of this task was to document the analytical evidence supporting the recommendations made for HealthConnect Clinic.

Recommendation 1: Provide Additional Support for Longer Booking Lead Times

Recommendation:

HealthConnect should consider stronger reminder and confirmation processes for appointments booked further in advance.

Supporting Evidence:

The analysis showed that no-show rates increased as booking lead time increased:

0-9 days: 29.01%

20-29 days: 44.44%

40-49 days: 61.62%

50-59 days: 67.92%

The pattern was also observed across appointment types and reminder-status groups.

Evidence-Based Rationale:

Appointments booked further in advance may benefit from additional patient reminders and confirmation closer to the appointment date.

Recommendation 2: Provide Targeted Support for Patients with Previous No-Shows

Recommendation:

HealthConnect should consider providing additional appointment support to patients with a history of missed appointments.

Supporting Evidence:

No-show rates generally increased with previous no-show history:

0 previous no-shows: 43.51%

1 previous no-show: 53.49%

2 previous no-shows: 59.36%

3 previous no-shows: 67.95%

The relationship remained visible across different appointment types.

Evidence-Based Rationale:

Previous no-show history can be used as an indicator for identifying patients who may benefit from additional confirmation, reminders, or other appropriate support.

Recommendation 3: Strengthen the Reminder Process

Recommendation:

HealthConnect should review and strengthen its reminder process to ensure eligible patients receive timely reminders.

Supporting Evidence:

The observed no-show rate was:

51.39% without a reminder

47.36% with a reminder

4.03 percentage-point difference

The lower no-show rate with reminders was also observed across appointment types and booking lead-time groups.

Evidence-Based Rationale:

The finding suggests that reminder status is associated with no-show behaviour and supports further review of the clinic's reminder process.

Overall Evidence

The Week 6 analysis provides evidence supporting three main areas of action:

1. Booking lead time can be used to identify appointments that may require additional attention.
2. Previous no-show history can help identify patients who may benefit from targeted support.
3. Reminder status provides evidence for reviewing and strengthening the existing reminder process.

These recommendations are based on observed relationships in the HealthConnect appointment data and the findings validated during Week 6.

Analytical Limitations

The evidence demonstrates associations rather than causation. Therefore, the recommendations should be treated as evidence-based actions for further implementation and evaluation rather than proof that any single intervention will directly reduce no-shows.

Small groups, particularly at very high booking lead times or with several previous no-shows, should also be interpreted cautiously.

Task 12: Cross-Track Analytical Contribution

Objective

The purpose of this task was to document how the Data Analytics work contributed to another project track during Week 6 and how the cross-track collaboration supported the wider HealthConnect project.

Track Collaborated With

Data Science

The Data Analytics track collaborated with the Data Science track to compare validated analytical findings with the no-show prediction model and its error analysis.

Analytical Contribution Provided

The Data Analytics track provided validated findings covering:

Booking lead time and no-show behaviour.

Previous no-show history and future no-show behaviour.

Appointment type and no-show behaviour.

Reminder status and no-show behaviour.

Relationships between booking lead time, previous no-shows and reminder status.

Validated Week 5 KPIs.

Segment-level findings relevant to model interpretation.

These findings were shared with Data Science to help assess whether observed patient and appointment patterns were consistent with the model's results.

Cross-Track Findings

The Data Science review showed strong alignment between the analytical findings and the model.

Booking lead time was identified as the model's strongest feature, accounting for approximately 57% of its predictive weight. This was consistent with the Data Analytics finding that no-show rates increased from 29.01% for 0-9 days to 67.92% for 50-59 days.

Historical no-show rate was identified as the model's second-strongest feature at approximately 12%, which was also consistent with the Data Analytics finding that no-show rates increased from 43.51% for patients with no previous no-shows to 67.95% for patients with three previous no-shows.

The approximately 4 percentage-point reminder gap observed by Data Analytics was also consistent with the Data Science review.

Contribution to Model Interpretation

The collaboration also helped clarify the Specialist Consultation finding.

Data Science had identified Specialist Consultation as having the highest model error rate at 42.2%. However, the Data Analytics findings showed that Specialist Consultation did not have the highest overall no-show rate. Follow-up had the highest overall no-show rate at 51.23%, compared with 47.44% for Specialist Consultation.

This distinction helped Data Science avoid interpreting the higher model error rate as evidence that Specialist Consultation inherently had the highest no-show risk.

The analysis instead highlighted the potential importance of interactions between appointment type, booking lead time and previous no-show history.

Integration Outcome

The cross-track collaboration resulted in the following improvements:

1. Data Analytics findings were reviewed against the Data Science model.
2. Booking lead time was independently confirmed as a strong predictive signal.
3. Historical no-show behaviour was independently confirmed as an important model feature.
4. The reminder finding was supported by the Data Science review.
5. The interpretation of the Specialist Consultation model error was refined.
6. The interaction between appointment type and booking lead time was identified as an area for further Week 7 investigation.
7. Data Analytics provided evidence that can support future modelling and testing decisions.

Evidence

The cross-track contribution is supported by:

Week 6 Data Analytics findings and validation tables.

Week 6 segment and relationship analyses.

Week 6 KPI validation.

Written feedback from the Data Science track.

Follow-up discussion regarding Specialist Consultation model error.

Analysis of appointment type $\times$ booking lead time.

Coordination records documenting the Data Analytics $\rightarrow$ Data Science integration.

Overall Contribution

The Data Analytics track contributed validated evidence that helped Data Science interpret and assess its no-show prediction model. The strong alignment between the analytical findings and important model features provided additional confidence in the relevance of booking lead time and previous no-show behaviour.

The collaboration also identified a potential interaction between appointment type and booking lead time for further investigation during Week 7.

This demonstrates that the Week 6 Data Analytics work contributed directly to cross-track integration, model interpretation and preparation for Week 7 testing, rather than operating as an independent analysis.

Week 7 Analytical Testing Requirements

Objective

The purpose of Week 7 testing will be to determine whether the Week 6 analytical findings, KPIs, recommendations and dashboard improvements remain reliable and useful when subjected to further testing and validation.

1. Validate Key No-Show Relationships

The following relationships should be tested further:

- Booking lead time and no-show behaviour.
- Previous no-show history and future no-show behaviour.
- Reminder status and no-show behaviour.

The purpose is to confirm that the relationships identified during Week 6 remain meaningful when tested further.

2. Test KPI Accuracy

The main KPIs should be rechecked to ensure that the calculations remain accurate after further development or changes to the report:

- Total Appointments
- No-Show Rate
- Attendance Rate
- Cancellation Rate
- Reminder No-Show Gap

3. Test Segment Consistency

The key findings should be tested across relevant appointment and patient segments to determine whether the observed patterns remain consistent.

Particular attention should be given to segments with small numbers of appointments.

4. Test Dashboard Functionality

The improved Week 6 dashboard should be tested to confirm that:

- Visuals display the correct measures.
- Filters and slicers behave as expected.
- KPI values remain accurate.
- The updated visuals continue to communicate the key findings clearly.

The Week 6 improvements do not introduce inconsistencies into the report.

5. Review Recommendations Against New Evidence

The recommendations developed during Week 6 should be reviewed against the results of further testing.

If new evidence changes the interpretation of a finding, the relevant recommendation should be revised accordingly.

6. Analytical Limitations to Monitor

Week 7 testing should continue to monitor:

Small sample sizes in some segments.

The possibility of misleading results from very small groups.

Whether observed relationships remain consistent.

The distinction between association and causation.

Expected Week 7 Outcome

By the end of Week 7, the Data Analytics work should have stronger evidence showing whether the Week 6 findings, KPIs, recommendations and dashboard remain reliable and suitable for the next stage of HealthConnect development.

Week 7 testing will focus on further validation and refinement rather than repeating the complete Week 5 or Week 6 analysis.